

AF-FORGE

Agentic Memory Architecture — Machine State Snapshot

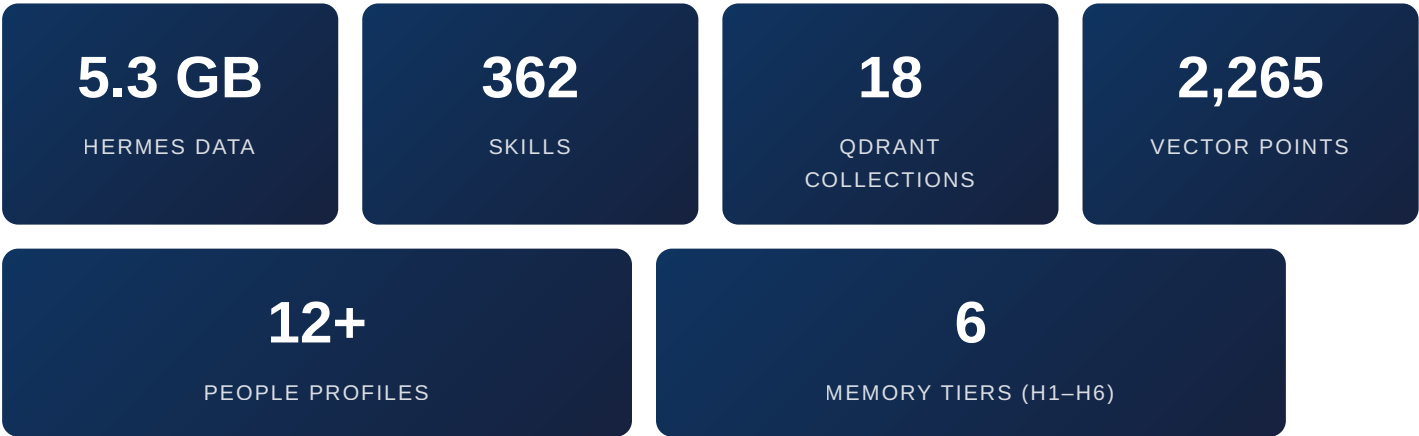
Host: forge | **Date:** 25 August 2026, 22:10 MYT

Operator: Muhammad Arif bin Fazil (F13 Sovereign)

Kernel: Linux 6.17.0-41-generic | **Uptime:** 33 days

DITEMPA BUKAN DIBERI — Forged, Not Given

1. Executive Summary



The AF-FORGE machine hosts the complete arifOS federation memory stack — a six-layer hierarchical system (H1–H6) backed by vector search (Qdrant), relational state (SQLite/Postgres), graph knowledge (Graphiti), immutable ledger (VAULT999), and session conversation history (Hermes sessions.db). Memory flows through a single kernel governor (`arif_memory`) with constitutional gating (F1–F13) on every write.

Key Insight: No organ writes to memory directly. All organs call the kernel's single interface, and the kernel fans out to vector, relational, and graph stores. VAULT999 (L6) is the terminal surface; the judge is always the court.

2. Machine State

Parameter	Value	Status
Hostname	forge	▲
Kernel	Linux 6.17.0-41-generic	▲
Uptime	33 days, 11 hours	● Stable
Disk (/dev/sda1)	309 GB / 387 GB (80%)	▲ High
RAM	21 GB / 31 GB used	Elevated
Swap	7.9 GB / 8 GB used	Critical
Load Average	2.61 / 3.20 / 3.34	Moderate

Swap Pressure: 99% swap utilization indicates sustained memory pressure. Consider adding RAM or reducing memory-hungry services. The 31 GB RAM is nearly fully committed with 9 GB in buff/cache.

3. Memory Architecture — Full Map

3.1 The Six Tiers (H-Series)

The arifOS memory hierarchy follows a six-tier model, each serving a distinct cognitive function:

Tier	Name	Function	Storage	Size
H1	Capture	Raw reality input — biographical truth, first-person data	Markdown files	18 KB
H2	Experience	Temporal event logs, monthly archives (2026-04 → 2026-08)	Monthly directories	~5 months
H3	Knowledge	Domain expertise — geoscience specs, ZKPC, figure pipelines	Markdown + images	592 KB
H4	Identity	Who Arif is — biographies, family, agent briefs	Markdown + PDF	372 KB
H5	Scars	Failure patterns — 30+ scar documents, scar registry	JSON + Markdown	280 KB
H6	Constitution	Scar roots, constitutional bedrock	Markdown	13 KB

3.2 Supporting Memory Surfaces

Surface	Path	Purpose	Size
People Registry	/root/memory/people/	Per-person dossiers (Abbah, Aliff, Syed, Nabilah, etc.)	88 KB
VVV Vault	/root/memory/VVV/	Void entries, vault.json — what was deliberately unmade	24 KB
Zen Helix	/root/memory/ZEN_HELIX.md	Cross-cutting zen doctrine synthesis	14 KB
Cross-Harness Manifest	/root/memory/CROSS_HARNESS_MANIFEST.md	Multi-agent protocol alignment	3.3 KB
Reality Memory	H1-capture/ ARIF_FAZIL_REALITY_MEMORY.md	Arif's full reality truth (18 KB)	18 KB

4. Vector Memory — Qdrant Collections

All vector embeddings use `bge-m3` (1024-d) unless noted. Qdrant runs at `127.0.0.1:6333`.

Collection	Points	Purpose	Write Gate
<code>petronas_knowledge</code>	1,460	PETRONAS corporate, Petros-Shell dispute, O&G intel	Domain-specific
<code>arifos_precedent</code>	255	PRL — sealed outcome precedents for decision constraint injection	Vault vectorizer
<code>arifOS_skill_mesh</code>	224	Skill embeddings for intent → skill routing	Skill catalog sync
<code>arifos_session_memory</code>	56	Session-scoped semantic memory	arif_memory (L3)
<code>arifos_memory</code>	49	Federation-wide semantic memory (core)	arif_memory (L3)
<code>federation_shared</code>	30	Cross-agent shared context	arif_memory (L3)
<code>arif_evidence</code>	16	Evidence trail for claims	Evidence discipline
<code>arifos_constitution</code>	14	Constitutional floor embeddings	Governance sync
<code>arifos_vault_canon</code>	13	Canonical vault entries	VAULT999 seal
<code>openclaw_memory</code>	9	OpenClaw agent memory	OpenClaw kernel
<code>arifos_vault_working</code>	7	Working vault (pre-seal)	arif_memory (L3)
<code>atlas333_eureka</code>	74	Atlas333 eureka/insight events	Eureka engine
<code>arifos_audio_memory</code>	2	Audio feature embeddings	Audio pipeline
<code>arifos_vault_ephemeral</code>	0	Temporary vault (cleared)	—
<code>arifos_vault_quarantine</code>	0	Quarantined (suspect) entries	—
<code>arifos_vault_sacred</code>	0	M3/M4 sacred — requires F13	F13 only
<code>mcp_capabilities</code>	0	MCP tool capability index	Registry sync
<code>test_memory_arch</code>	—	Test collection (development)	—

18 collections, 2,265 total points. The heaviest collection is `petronas_knowledge` (1,460 pts — 64.4% of all vectors), reflecting deep PETRONAS/Petros/Shell intelligence accumulation. The precedent collection (`arifos_precedent`, 255 pts) feeds the PRL (Precedent Recall Layer) which injects structural constraints before reasoning.

5. State Databases

Database	Path	Size	Purpose
Hermes State	<code>~/.hermes/state.db</code>	286 MB	Primary Hermes agent state — all runtime data
Sessions DB	<code>~/.hermes/sessions/sessions.db</code>	0 bytes	Session index (empty — sessions via JSON)
Sessions JSON	<code>~/.hermes/sessions/sessions.json</code>	45 KB	Session metadata store
Request Dumps	<code>~/.hermes/sessions/</code>	~117 MB	44 request dump files (Aug 4–25)
Cron Executions	<code>~/.hermes/cron/executions.db</code>	348 KB	Cron job execution history
Cron Jobs	<code>~/.hermes/cron/jobs.json</code>	59 KB	2 active cron jobs
Notepad	<code>~/.hermes/cron/notepad.db</code>	12 KB	Cron scratchpad
Kanban	<code>~/.hermes/kanban.db</code>	—	Task board state
Verification Evidence	<code>~/.hermes/verification_evidence.db</code>	—	Work verification audit trail
GEOX Megamemory	<code>/root/GEOX/.megamemory/knowledge.db</code>	84 KB	Geoscience domain knowledge
WEALTH Megamemory	<code>/root/WEALTH/.megamemory/knowledge.db</code>	84 KB	Capital/trading domain knowledge
Hermes State Snapshots	<code>~/.hermes/state-snapshots/</code>	1 snapshot	Pre-update backup (2026-08-14)

Sessions DB Anomaly: `sessions.db` is 0 bytes while `sessions.json` (45 KB) holds session metadata. Request dumps (117 MB across 44 files) carry the actual conversation payloads. This is a non-standard pattern — the SQLite index appears unused.

6. Hermes Profiles

Profile	Purpose	Has Own Skills?
default	Primary — Arif's main agent (this session)	Yes
aaa-hermes	Federation AAA integration profile	Yes
hermes_forge	A-FORGE dedicated execution profile	Yes
hermes_apex	APEX constitutional governance profile	Yes
hermes_asi	ASI (artificial superintelligence) research profile	Yes
nabilah	Nabilah — English teacher agent (cron: RPH 6:30AM)	Yes

7. Organ Memory Surfaces

Each federation organ maintains its own memory footprint alongside the shared Qdrant/VAULT999 infrastructure:

Organ	Repo Path	Megamemory	Knowledge DB	Special Memory
AAA	/root/AAA/ (1.1 GB)	—	—	Knowledge graph (5 files, 122 KB), 26 doctrine files, 90+ governance docs
A-FORGE	/root/A-FORGE/	—	—	VAULT999 ledger (read-only), forge_work artifacts
GEOX	/root/GEOX/	.megamemory/	84 KB	Geoscience knowledge base
WEALTH	/root/WEALTH/	.megamemory/	84 KB	Capital markets knowledge base
WELL	/root/WELL/	—	—	Wellness/homeostasis data

8. Ephemeral Layer — Redis

Metric	Value
Used Memory	1.87 MB
Fragmentation Ratio	3.86 (high — consider <code>redis-cli memory purge</code>)
Purpose	Session cache, rate limiting, ephemeral state

9. Skills Memory — Procedural Knowledge

362 skills across the skill directory, organized by naming convention into functional domains:

Prefix/Domain	Example Skills	Count (est.)
AAA-*	OCR, ASR, Audio EMD, Voice Cloning, TTS Catalog	~12
AGI-*	Agentic Web, Audio Quantum, Dream Engine, Plan DAG	~12
ASI-*	Agent Invariants, Drift Watch, Fabrication Prevention	~7
APEX-*	FFF Loop, Quantum Eureka	~2
AUDIT-*	Skill Mesh, Drift Detector, Recursive Audit, Skill Atlas	~4
FORGE-*	CI/CD, Docker, FastAPI, GitHub, MCP, PR Governance, VPS	~50+
FLAME-*	Operator, Router	~2
EUREKA777-*	Paradox Resolution	~1
RSI-*	Federation Mesh, Recursive Improvement	~2
Domain-specific	Geology, Trading, Legal, Biohacking, Media, Social	~50+
Workflow/Core	Incident, Issue Triage, Subagent, Cross-agent	~20+
Substrate/Primitives	Memory, Observe, Think, Verify, Forge, Judge	~30+
Drafts (_drafts)	Event-captured workflow fragments	~13

10. Governance & Doctrine Memory

Layer	Count	Path
Canonical Instructions	26 files	/root/AAA/instructions/
Governance Documents	90+ files	/root/AAA/governance/
Prompts	10 files	/root/AAA/prompts/
Sovereign Patches	4 files	/root/docs/sovereign-patches/
Carry Forward Backups	36 files	~/.local/share/arifos/carry_forward_backups/
Carry Forward (current)	1 file	~/.local/share/arifos/carry_forward.json

Carry Forward: The active carry forward traces to session SEAL-333-AGI-20260825-gemini (closed 2026-08-25T17:48:00Z). It tracks 5 completed items and 4 open loops including F1 hook overreach and OAuth bridge gaps.

11. VAULT999 — Immutable Ledger

VAULT999 is the terminal memory surface — the immutable, hash-chained receipt log. Currently at `/root/AAA/vault999/VAULT999.md` (411 bytes). The ledger tracks sealed decisions with cryptographic receipts.

The `arifos_precedent` Qdrant collection (255 points) is the vectorized shadow of VAULT999 — every sealed outcome gets embedded and indexed for PRL (Precedent Recall Layer) injection. PRL injects structural constraints at `mind_reason.py` line ~348 before reasoning begins, gated by dual-filter: `blast_radius match +  $\Omega_0$  ambiguity failsafe`.

12. Knowledge Graph — Domain Intelligence

File	Size	Content
<code>visual-intelligence-map.md</code>	56 KB	Visual processing capability map
<code>video-intelligence-map.md</code>	27 KB	Video processing pipeline
<code>audio-intelligence-map.md</code>	19 KB	Audio cognition architecture
<code>hermes-epistemic-architecture.md</code>	13 KB	Epistemic layer design
<code>hermes-k2-contrast.md</code>	7 KB	K2 contrast analysis
<code>voice-sample/</code>	5 files	i-ARIF voice samples (Tukki presets)

13. People Memory — Human Network

Person	Directory	Notes
ARIF	people/ARIF/	Self — sovereign, F13
SYED	people/SYED/ (8 subdirs)	Abang sado — KL, relay info. MOTS-c/NAD+/CoQ10 stack. BANK: 8755310538
NABILAH	people/NABILAH/	Eng teacher, cron 6:30AM RPH agent
MAK	people/MAK/	Mother
ABBAH	people/ABBAH/	Father
FAZIL	people/FAZIL/	Family member
FARIDAH	people/FARIDAH/	Family member
ALIFF	people/ALIFF/	Family member
IZZU	people/IZZU/	Family member
WAWA	people/WAWA/	Telegram alias — DM source
FAMILY	people/FAMILY/	Family constellation overview
SCAR-WITNESSES	people/SCAR-WITNESSES/	People tied to scar events

14. Scar Registry — Failure Memory

30+ scar documents + H5_SCAR_REGISTRY.json (30 KB) tracking constitutional failures and lessons learned:

Category	Scars	Key Learning
Shadow/Syed	SCAR_SYED_SHADOW_V2-V9	Shadow reading protocol evolved through 8 iterations (26 KB cumulative)
Fabrication	SCAR_NOTEBOOKLM_EXIT_FABRICATION	Never fabricate source provenance
Memory Doctrine	SCAR_MEMORY_DOCTRINE_V1	Memory write discipline
Institutional	SCAR_NUKLEAR_LETTER, SCAR_SB412	Institutional pattern recognition
Identity	SCAR_SIBLING_CONSTELLATION	Family relationship boundaries

Compression	SCAR_AKUSAJACUKUP, SCAR_BEKANTAN	Over-compression risks
Whistleblowing	SCAR_MAKCIKGPT	Public speech boundaries
Numbered (1–11)	SCAR-1 through SCAR-11	Foundational operational failures

15. Cron & Scheduled Memory

Metric	Value
Active Cron Jobs	2
Execution History DB	348 KB
Jobs Config	59 KB (jobs.json)
Backup Configs	12 historical backups
Notepad	12 KB scratchpad
Output Directories	23 cron output folders

16. Memory Flow — How Data Moves

Write Path (Store)

INTAKE

User message / tool output / agent observation →
`arif_memory(mode="store")`

L3 Qdrant

Required. bge-m3 embedding → `arifos_memory` collection. Failure =
stored=False.

L4 Postgres

Durable relational store. Warns on failure, does not block.

L5 Graphiti

Graph knowledge. Advisory only. Never auto-woken (888 owns).

L6 VAULT999

Immutable ledger. Sealed receipts. Hash-chained. Read-only after seal.

Read Path (Recall)

QUERY

Agent needs context → `arif_memory(mode="recall")`

PRL Gate

Precedent Recall Layer intercepts → dual-gate (blast_radius + Ω_0 ambiguity)

Qdrant Search

Cosine similarity search across relevant collections

HOLD Gate

If $P(\text{PRL}) < 0.95$ or ambiguity detected → F1 HOLD. Human decides.

REASONING

[PRL CONSTRAINT] block injected at END of reasoning prompt (recency-bias
hardening)

17. Critical Observations & Recommendations

Swap Critical: 7.9 GB / 8 GB swap used. This is the #1 operational risk. Every memory-intensive operation (Qdrant search, session loads, cron runs) will page-fault heavily. Recommend: add RAM or reduce concurrent memory consumers.

Disk at 80%: 309 GB / 387 GB. Request dumps alone consume ~117 MB and grow with each session. Consider archiving old dumps (pre-Aug-15) and running `docker system prune`.

Sessions DB Empty: `sessions.db` is 0 bytes. Session queries via `session_search` may be reading from the JSON file or a different backend. Verify this is intentional.

Qdrant Healthy: 18 collections, all responsive. `petronas_knowledge` is the dominant vector store (64.4%). Precedent collection at 255 points feeds PRL correctly.

Memory Hierarchy Intact: All 6 tiers (H1–H6) present and populated. People registry covers 12+ profiles. Scar registry carries 30+ lessons across 5 evolution families.

Redis Fragmentation: 3.86x fragmentation ratio on only 1.87 MB. Not critical at current scale but indicates memory allocation pattern issues. `redis-cli memory purge` recommended.